

IndEgo: A Dataset of Industrial Scenarios and Collaborative Work for Egocentric Assistants

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<https://indego-dataset.github.io/>

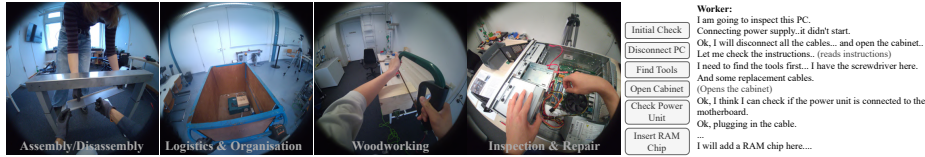


Fig. 1: Some examples from the IndEgo dataset showing different industrial cases. **a.** Assembly/Disassembly and Collaborative Work (further elaboration on Figure 2), **b.** Logistics and Organisation, **c.** Woodworking, **d.** Inspection and Repair (The worker's narration and the annotated actions are also shown).

Abstract. We introduce IndEgo, a multimodal egocentric and exocentric dataset addressing common industrial tasks, including assembly, disassembly, logistics and organisation, inspection and repair, woodworking, and others. The dataset contains 3,460 egocentric recordings (approximately 197 hours), along with 1,092 exocentric recordings (approximately 97 hours). A key focus of the dataset is collaborative work, where two workers jointly perform cognitively and physically intensive tasks. The egocentric recordings include rich multimodal data and added context via eye gaze, narration, sound, motion, and others. We provide detailed annotations (actions, summaries, mistake annotations, narrations), metadata, processed outputs (eye gaze, hand pose, semi-dense point cloud), and benchmarks on procedural and non-procedural task understanding, Mistake Detection, and reasoning-based Question Answering. Baseline evaluations for Mistake Detection, Question Answering and

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collaborative task understanding show that the dataset presents a challenge for the state-of-the-art multimodal models. Our dataset is available at: <https://huggingface.co/datasets/FraunhoferIPK/IndEgo>

Keywords: Egocentric Vision · Industrial Scenarios · Dataset · Mistake Detection · Video Question Answering · Summarisation

1 Introduction

Egocentric Vision and AI are gaining significant attention, driven by both their economic potential and the recent development of general-purpose foundational models [14, 27, 40, 58]. One of the key goals of this field is to develop helpful AI assistants that understand user’s actions, intentions, and needs, and provide valuable guidance [14, 45]. Such assistants would be valuable in learning and acquiring new skills, navigating new environments, and improving user experience. Additionally, egocentric research is also relevant to the development of embodied agents that can learn from demonstrations, assist and collaborate with humans [2, 6, 47, 49, 59].

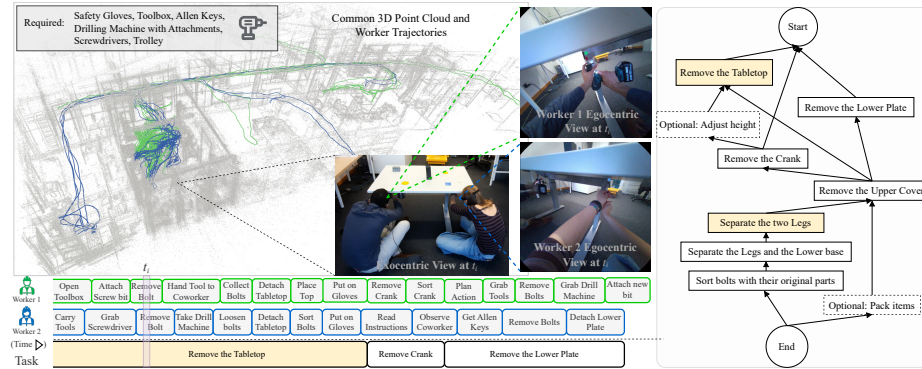


Fig. 2: A scenario of a *disassembly* process from the IndEgo dataset. The two participants work collaboratively on the task. The semi-dense point cloud and the user trajectories are generated by processing the raw data from the Aria device [14, 15]. The egocentric perspective of the two participants with the projected eye gaze point can be seen in relation to the 3D environment and the exocentric view. **Bottom:** The annotations from each worker’s perspective, and the keysteps in the process. **Right:** The corresponding task graph for the procedure. The flow of activities is from top to bottom, and dependencies are shown with an arrow. denotes labour-intensive steps.

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To enable and accelerate research in these areas, several datasets and benchmarks have been introduced [12, 21, 27]. Current datasets and initiatives focus heavily on daily activities and procedures [12, 21, 22]. This is in part because these are applicable across different cultures and lifestyles, and also because it is relatively easier to design experiments and collect data in settings such as the kitchen [12, 26, 44]. Some recent datasets also focus on industry-like contexts [48, 51], however, the datasets published in true industrial settings remain low. Industrial scenarios offer a rich domain for egocentric vision research due to their complexity, diversity of tasks, and dynamic environments [48, 51]. Workers in industrial settings perform intricate manual operations, manipulate various tools and components, and navigate in cluttered and unpredictable spaces [10, 53]. These conditions present unique challenges for egocentric vision systems, such as accurately recognising actions and gestures in real-time, identifying and localising tools and parts amidst visual occlusions, and adapting to varying lighting and environmental conditions. These scenarios are underrepresented in current egocentric vision datasets and research.

Egocentric Vision-based assistive technologies can offer several unique and practical implications for improving productivity, safety, and efficiency in industrial operations. To investigate the future application potential of wearable egocentric vision-based assistants, we present the **IndEgo** dataset, consisting of diverse Industrial tasks and processes from an Egocentric perspective, with a supplementary Exocentric view for reference in several cases. IndEgo comprises 3,460 unique egocentric video recordings (totalling 197.1 hours), and 1,092 (totalling 96.8 hours) accompanying exocentric video recordings. The dataset includes scripted and unscripted actions from participants in a dedicated industrial facility, collected in diverse conditions (lighting, time of the day, setup). Figure 1 shows an example. The dataset also includes collaborative work, where two participants work together on a common task in various roles (partners, teacher-student, leader-assistant). AI assistants and embodied agents of the near future will work alongside other humans on such tasks, which makes this a relevant subdomain for further exploration and research.

2 Related Work

AI Assistants. The mainstream adoption of AI technology has led to a strong focus on the development of digital assistants that can chat (text and audio), answer questions, understand images, videos, and process multimodal data [32, 40, 42]. Research on embodied agents that can manipulate their environment and assist humans is also accelerating [19, 50, 55, 56].

Egocentric Computer Vision. The goal is to understand the world from a human-centric perspective [17, 27, 36]. Users often collect data by wearing a portable camera device (or other sensors) and performing various activities [12, 21]. This provides insight into several scenarios that cannot be captured by traditional vision research [21, 45]. This introduces several novel challenges, due to the dynamic nature of the scenes and environments [14, 17, 18]. Research

directions include video understanding and summarisation [13,17,28], hand-object interaction [4,5], interpersonal interactions [16,35], among others. Additionally, eyewear-based sensors also introduce novel research opportunities on eye-gaze understanding and prediction [23,30,31]. Smart glasses with camera and audio have recently seen commercial success [37], along with continued interest in AR/VR technology [1,38].

Egocentric Datasets. Several open-source datasets and benchmarks have recently been introduced, including EPIC-KITCHENS [12] and Ego4D [21]. These datasets have highlighted the need for research on audiovisual diarization, episodic memory, action anticipation, localisation and tracking, and other domains. A large fraction of the published datasets focus on everyday activities, and indoor and outdoor tasks [12,21,33]. Recent datasets such as CharadesEgo [52], EgoExo4D [22], and EgoExoLearn [24] also utilise exocentric/allocentric perspectives, with a goal of improving understanding, enabling cross-view representation learning and skill transfer. Some industry-specific tasks, such as assembly-disassembly, goal step understanding, and skill assessment, have also been proposed [22,51]. Benchmarking tasks, such as mistake detection, have also seen increased attention [26,44,51].

Gaps and Opportunities. In this paper, we study and present industrial contexts and potential cases in which an egocentric vision-based assistant can be expected to operate in the near future. Such an assistant would be required to understand the worker’s surroundings and their personal context, guide the user through complicated environments and tasks, answer questions, detect mistakes, and reason about various factors with careful consideration [9,14,41,45]. Datasets such as Meccano [48], Assembly101 [51], and HoloAssist [60] involve industry-like situations, with participants carrying out procedural tasks and following instructions. However, most of these tasks involve participants working at a workstation or a desk [51]. In real-world scenarios, the worker is needed to move from one place to the other, carry out long and arduous tasks, and collaborate with others. The recent introduction of the Project Aria Toolkit unlocks several opportunities for data collection and research [14,43]. The Aria device is lightweight and mimics the form factor that future egocentric wearable devices are likely to have [14,38]. This enables the users to move freely and capture context-rich multimodal data in their environment. Thus, participants can perform cognitively and physically demanding tasks, similar to a true industrial setup, while being able to move naturally and untethered from a working desk [8]. Secondly, the toolkit also allows users to work collaboratively on a given task, which can be studied in much greater detail via the multimodal sensor output. Figure 2 shows an example. Lastly, prior datasets focus on short to medium tasks [51,60], leaving long-horizon activities (20+ min) underrepresented.

3 IndEgo Dataset

Industrial Scenarios. We categorise tasks into five broad groups: *assembly and disassembly*, *inspection and repair*, *logistics and organisation*, *woodworking*, and

miscellaneous (others). There may be overlap, especially in longer tasks (e.g., the worker transports the device and tools to another location before disassembly); in such instances, we defer to the overall goal of the video for categorisation. The tasks are cognitively and physically demanding, requiring skill and effort in terms of reasoning and manipulation.

Tools, Devices, and Machines. The tools used for the work include common and domain-specific tools found in industry or research labs. For assembly/disassembly, we use mechanical devices, PC cabinets, and proprietary assemblies.

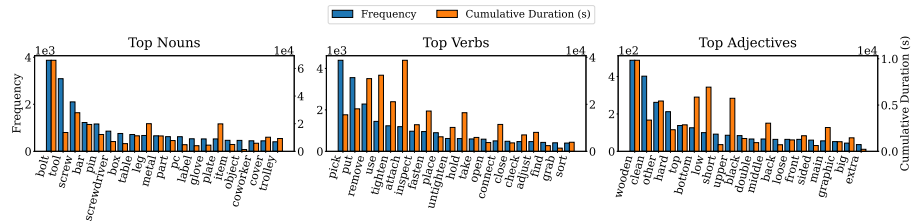


Fig. 3: Grouped bar charts of frequencies (left axis) and durations (right axis) for the fine-grained action annotations: Top 20 nouns (left), verbs (middle), and adjectives (right). Our dataset covers diverse industrial contexts, which are not represented by current egocentric/exocentric datasets. This highlights the multimodality and human-centric attributes of IndEgo.

Participants. The data was collected by 20 participants, 15 male and 5 female. The selection was done in an unbiased manner, and the ratio reflects the natural skew seen in industrial work. They have varying degrees of experience in industrial skilled work (beginner to expert), and come from different nationalities and ethnic backgrounds. For our study, all participants used only English for demonstration and narration.

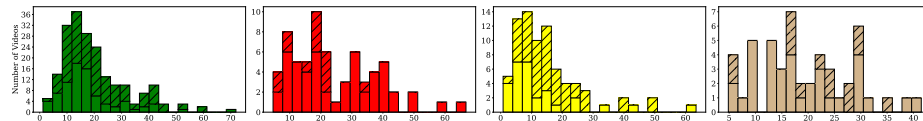


Fig. 4: Histogram of egocentric video durations (minutes) of medium-longer task sequences for each category: assembly/disassembly, inspection/repair, logistics/organisation, and woodworking. The hatched regions represent two-person collaborative tasks.

Category	T _{avg}	1-Person	Collaborative	Narration	#Ego	T _{Ego} (h)	#Exo	T _{Exo} (h)
Assembly	15.2 m	✓	✓	✓	188	47.5	152	30.4
Disassembly	11.1 m	✓	✓	✓	136	24.9	112	17.0
Inspection and Repair	7.8 m	✓	✓	✓	238	30.9	202	17.7
Logistics/Organisation	4.5 m	✓	✓	✓	456	35.4	158	8.1
Woodworking	7.5 m	✓	✓	✓	148	18.4	116	14.9
Miscellaneous	1.5 m	✓	✓	✗	378	9.4	352	8.7
Tools/Objects in Context	120 s	✓	✗	✗	604	20.1	–	–
Tools/Objects Demo	53 s	✓	✗	✓	302	4.5	–	–
Singular Actions	21 s	✓	✓	✗	1010	5.9	–	–
Total	205 s	✓	✓	✓	3460	197.1	1092	96.8

Table 1: A breakdown of the IndEgo dataset, showing the key categories and related statistics. T_{avg} gives the average duration of the recordings, #Ego gives the number of videos from the Egocentric perspective, T_{Ego} gives the total cumulative time for egocentric data, #Exo gives the number of videos from the fixed exocentric perspective, T_{Exo} gives the total cumulative time for exocentric data.

3.1 Data Collection Process

Hardware Setup. Each participant used the Aria device [14] for the egocentric perspective. We defined a custom profile and recorded data from all available sensors for all our sessions. The camera sensors recorded data at 10FPS. The main RGB camera was set to the highest resolution (8MP: 2880 × 2880). For longer recordings, we use a portable powerbank, connected via a USB cable. With this setup, the maximum duration of a single recording was approx. 68 min (due to the on-device storage). Additionally, we use an external exocentric camera for several (1092) instances. We use different camera sensors (Sony A6400 APSC with Sigma 16 and 30 mm lens, Samsung Galaxy A51 and iPhone 16 as backup) with 1920 × 1080 resolution. The external camera is placed in the approximate position of a hypothetical third-person human observer, with a clear view of the participant and the working area. The raw data collected by the participants was stored and processed on a dedicated Workstation.

Privacy and Ethics. The participants were thoroughly informed about the general scope of the project and the intention to publish and open source the dataset. After obtaining written consent, they were given a primer on using the Aria device and the standard practices (work and safety, ethics, organisation of used tools, etc.). The working space is shared with other researchers, technicians and workers, who were all informed about the ongoing recording activities and protocol to avoid breach of their privacy. Written consent was also obtained from those who appeared in the background in the recordings. The project objectives, collected data, and other contents were examined by the reviewing authorities and approved for this submission.

Protocol. The activities took place at an industrial research facility over several months. Each session starts with a review of the planned tasks, followed by gathering the necessary tools/devices. Each participant calibrated the Aria

device to accurately track their eye gaze (via the eye tracking cameras). The different tasks and scenarios were carried out in different settings: First, the participants were given a goal (e.g. *disassemble the mechanical frame*) without any specific steps or instructions. They were asked to carry out the task by narrating their inner monologue, including instances when they were not sure about a certain action or when they had made a mistake. Second, they received thorough, step-by-step instructions, as well as guidelines for confirming whether they had executed the task correctly. Thirdly, they were accompanied by a fellow participant for challenging activities where they collaborated on the task as a team. In the latter, both participants wore the Aria device and were asked to talk amongst each other and narrate their thoughts. We also organised scenarios where the participants had two identical setups and tools, and the goal was for the more experienced person to guide and teach the other person through the process via demonstration and dialogue. The participants had access to an attendant, who intervened only in cases of emergency or when the participants requested guidance. Lastly, the participants repeated certain sequences from the dataset with *mistakes* in various scenarios for the *Mistake Detection* (MD) benchmark (elaborated in Section 4).

They also collected data on *singular actions* under different circumstances. These are general and shorter actions with one specific goal (e.g. unloading a trolley, fastening a bolt, clamping a wooden block) which are likely to occur in different contexts in the industry. To further add variation, participants employ different objects (heavy/light, single/many), locations, and backgrounds. An additional set of recordings was done with the *tools and objects in context*, with the users interacting with/using the tools and objects. The focus is on the particular tool (w.r.t. eye gaze, hands, etc.) in various circumstances. Finally, the participants recorded explanations and demonstrations (with narration) on how to use the different tools and their purpose. Table 1 gives a breakdown of the various segments of the dataset.

3.2 Annotation, Labelling, and Data Modalities

The participants annotate the data for their own recordings, which is then reviewed by a second person. For collaborative work, each user annotates their individual actions. Figure 2 shows an example, where the ongoing procedure, and the actions of the two workers, is put into context. The actions are annotated into logical segments, e.g. *connect metal bar*, *attach drill bit to the drilling machine*, etc. We also annotate the keysteps for the procedural activities.

We share the raw recorded data from all sensors, along with the processed video files and the extracted frames. The processed outputs from the Aria recordings include eye gaze estimation, hand pose estimation, semi-dense point cloud and trajectory data. For procedural activities, the task graphs, instruction guides (originally made available to the participants) and metadata are available. Additionally, we provide extracted audio transcripts of the narrations and dialogue.

Inter Annotator Agreement. We conducted a brief study on a portion of the dataset (cumulative 3 hours of data, all scenarios). The recordings were

Dataset	Scenario	Hours (Ego)	Exo	Collaboration	Gaze	Motion	Narration	Actions	Keysteps	Mistakes	QA
EPIC-KITCHENS [12]	Kitchen	100	✗	✗	✓	✗	✓	✓	✗	✗	✗
CharadesEgo [52]	Daily	34	✓	✗	✓	✗	✓	✗	✗	✗	✗
Ego4D [21]	Multiple	3670	✓	✗	✗	✓	✓	✓	✓	✗	✗
LEMMA [25]	Daily	10	✓	✓	✗	✗	✗	✓	✗	✗	✗
Ego-Exo4D [22]	Multiple	221	✓	✓	✗	✓	✓	✓	✓	✗	✗
EgoExoLearn [24]	Daily, Lab	120	✓	✗	✓	✗	✓	✓	✓	✗	✗
Nymeria [34]	Daily	300	✓	✓	✓	✓	✓	✓	✗	✗	✗
AssistQ [60]	Assistive	3	✗	✗	✗	✓	✓	✓	✗	✗	✓
Meccano [48]	Industry-like	7	✗	✗	✓	✗	✓	✓	✗	✗	✗
HoloAssist [60]	Assistive	166	✗	✓	✓	✓	✓	✓	✓	✓	✗
Assembly101 [51]	Industry-like	42	✓	✗	✗	✗	✗	✓	✓	✓	✗
IndEgo (ours)	Industrial	197	✓	✓	✓	✓	✓	✓	✓	✓	✓

Table 2: Comparison with related datasets on scenarios, modalities, and annotations. Hours (Ego) refers to the cumulative duration of distinct egocentric video recordings. **Top:** Datasets with diverse and everyday scenarios. **Bottom:** Datasets in an industry-like or assistive setting.

annotated by up to 3 participants separately. For each annotation, we assess the agreement between the temporal annotations in each frame.

Keystep Annotations: We see an excellent agreement between the annotators, with a Krippendorff’s Alpha of $\alpha = 0.97$. For these, the only divergence in agreement tended to be the start and end of a defined step (e.g. 210s vs 212s in a recording of 557s in total).

Finegrained Annotations: For a strict agreement (exact lemmatised verb + noun match for a given frame), we report a Krippendorff’s Alpha of $\alpha = 0.25$. However, upon grouping similar verbs and nouns together (e.g. [detach, remove], [allen wrench, hex key], [get, grab, pick up], [colleague, coworker]), the value rises to $\alpha = 0.54$. We acknowledge there are certain variances between the way in which participants annotate and break up their actions, e.g. one participant annotated a temporal segment as *pick up device*, followed by *rotate device*, while another participant annotated the entire segment as *flip device*. We also see variance in the way the annotators describe an action, e.g. *search the drawer* and *find tool* were annotated for the same action. Similarly, *hold metal bar* and *assist coworker* were also annotated for the same action in a collaborative task. We believe that such differences are not a weakness, but rather reflect different ways in which an action may be interpreted.

3.3 Dataset Distribution and Analysis

The dataset comprises 197.1 hours of egocentric data, equating to 7.1M frames from the main RGB camera of the Aria device, alongside 13.9M SLAM frames and 13.9M eye-tracking frames, synchronised with the main RGB recordings. Additionally, the dataset includes 96.8 hours of exocentric recordings, corresponding to 10.5M RGB frames. Table 1 provides a detailed breakdown of the categories, including Mistake Detection (MD) tasks, which are generally of shorter duration, as well as medium-to-long industrial tasks. A further breakdown of these longer sequences is illustrated in Figure 4, which also highlights the distribution of collaborative recordings across four task categories. The miscellaneous category,

primarily consisting of MD recordings, includes tasks that do not clearly fit within the other predefined categories.

Participants provided annotations for approximately 34k fine-grained actions in total, each containing task-specific verbs, nouns, and adjectives. Figure 3 visualises the Part-of-Speech (POS) breakdown across medium-to-long task sequences. The frequency of the top occurring POS and their cumulative duration across all categories is also shown. The median annotation duration is 5.7 seconds, with a minimum duration of 0.19s (e.g. *hand object over to coworker*) and a maximum of 214s (e.g. *observe demonstration from coworker*). Users also labelled key procedural steps for each sequence, providing structured keystep annotations to support task understanding and benchmarking.

3.4 Comparison with Current Datasets

Table 2 gives an overview of the attributes of IndEgo in relation to other egocentric datasets. The bottom portion shows datasets with a primary focus on assistive technologies and industry-like scenarios. Our dataset aims to fill a crucial gap w.r.t. industrial cases, collaborative work, as well as physically and cognitively demanding tasks. Moreover, its multimodality (video, audio, narration, gaze) adds to the research and application potential.

Comparison with Project Aria-related datasets. Multiple datasets collected using the Project Aria device and toolkit have been released, including Ego-Exo4D [22], Aria Everyday Activities [33], Aria Digital Twins [43], Aria Synthetic Environments [46], HOT3D [5] and Nymeria [34]. All these datasets harness the multimodal sensor capabilities and the data processing tools available. Our dataset is unique in terms of its focus on industrial cases and assistance scenarios.

4 Benchmarks and Evaluation

A dedicated workstation (Nvidia A6000 GPU) is used for all experiments. We use consistent hyperparameters for all models. Abbreviations for the models used; VL3: VideoLLaMA3 8B [62], IVL2.5: Intern-VL-2.5 (InternViT-300M-448px-V2_5) [11], QVL2.5: Qwen2.5-VL-7B [3], GFT: Gemini 2.0 Flash Thinking Experimental [20]. * via API, Experimental, model architecture unspecified. All models use 480p video@10FPS.

4.1 Mistake Detection.

We recorded 1166 egocentric sequences across 25 tasks and all five categories, with correct actions and *planned + unplanned mistakes*. These involve procedural tasks (e.g. *repairing a PC*) as well as non-procedural tasks (e.g. *tidying the workspace after finishing work*). The tasks include at least 42 recordings each, with an average of 18 correct and 24 with *mistakes*. These involve common types of errors, such as *skipping* a procedural step, doing things in the *incorrect*

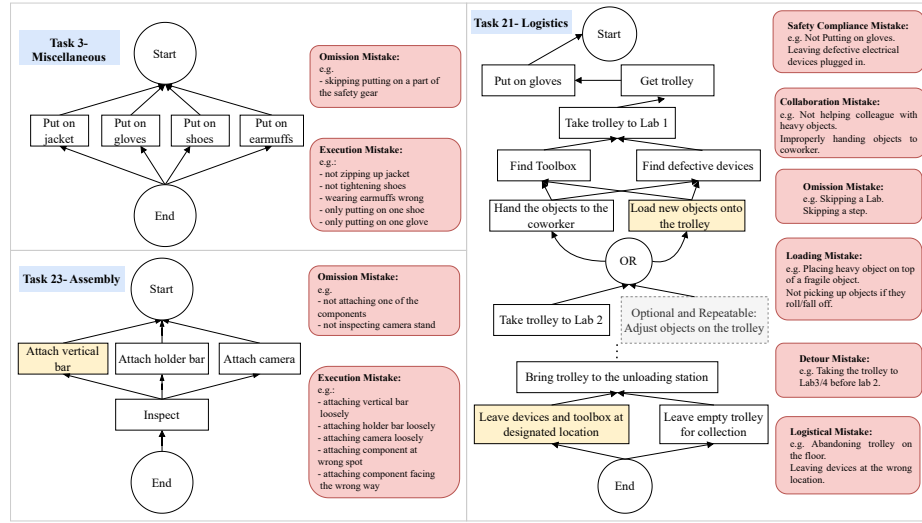


Fig. 5: Task graphs for some scenarios from the MD benchmark, along with some commonly seen mistakes (not exhaustive). The flow of actions is from top to bottom, and dependencies are shown with an arrow. denotes labour-intensive steps. **Left:** *Miscellaneous* sequence #3 at the top, and *Assembly* sequence #23 at bottom. **Right:** *Logistics* sequence #21. The graph is shortened due to spatial constraints.

order, and adding unnecessary steps. We also add *industry-specific mistakes*, such as not following workplace safety guidelines, mishandling tools/devices, and misplacing items. Additionally, for collaborative tasks, we include mistakes such as carelessly handing objects to a coworker and not helping a coworker carry out a physically strenuous task. Figure 5 shows an example of a collaborative task, with some examples of mistakes. We provide exocentric recordings for most tasks. We find that the exocentric perspective can provide additional context for the steps in case of limited egocentric visibility (e.g. *wearing PPE/safety kit*). For annotations, we provide the step-wise action segments and mistake labels, along with the descriptions of the mistakes. We also provide task graphs for the correct sequences.

In industrial settings, certain mistakes can be more consequential/critical than others. Hence, we add context-dependent categorisation for such errors. These include Severe Mistakes (S) - which can lead to unnecessary disruptions (e.g. mishandling a fragile object); mistakes leading to failure of the entire process (PF) - e.g. placing the wrong item in a shipment; mistakes that impact all future steps (IF) - e.g. forgetting to open the trolley hatch before loading material; and mistakes posing a risk of physical harm to self or other (H) - e.g. cleaning a wound with a dirty cloth. The majority of mistakes in the dataset do not fall into either category; 2.3% are S, 18.7% are PF, 7% are IF, and 5% are H.

Table 3 gives baseline results on the MD benchmark. For zero-shot evaluation, the VLMs are presented with the video and asked to predict whether it contains

	Approach	P	R	F1	F1 ^S	F1 ^{PF}	F1 ^{IF}	F1 ^H
ZS	VL3 [62]	15.6	46.2	23.3	36.2	38.2	27.4	32.1
	IVL2.5 [11]	16.2	48.2	24.2	38.1	37.1	29.0	33.2
	QVL2.5 [3]	15.9	50.1	24.1	38.8	36.5	28.8	34.1
	GFT* [20]	35.6	48.2	40.9	51.2	42.2	34.7	48.0
MLP	VL3 [62]	30.4	56.7	39.5	48.1	38.8	32.1	41.3
	IVL2.5 [11]	31.6	50.0	38.7	47.7	39.1	30.5	42.2
	QVL2.5 [3]	31.4	51.6	39.1	42.6	39.8	35.4	44.0
Tr	VL3 [62]	34.5	33.3	33.9	39.2	35.5	29.1	38.5
	IVL2.5 [11]	30.1	41.7	35.5	36.5	38.7	32.1	39.2
	QVL2.5 [3]	33.3	41.0	36.7	37.0	39.4	29.5	36.7
MLP	VL3 [62] (EM)	21.3	55.0	30.7	36.2	38.2	30.1	32.2
	IVL2.5 [11] (EM)	23.3	49.2	31.6	35.2	32.7	31.6	30.5
	QVL2.5 [3] (EM)	24.1	51.0	32.7	34.2	32.0	32.1	40.1

Table 3: Baseline Results for Mistake Detection. **Top:** Zero-shot evaluation - the F1^S and F1^H scores are higher because the model was prompted for the particular mistake. **Middle:** Finetuning binary classification (Mistake/Correct) via an MLP. and Transformer. **Bottom:** Early Mistake Detection (EM), where only 50% of the initial frames of the segment are available for the prediction.

	Approach	P	R	F1	F1 ^S	F1 ^{PF}	F1 ^{IF}	F1 ^H
Ego	VL3 [62]	17.1	48.0	25.2	34.1	37.2	28.4	35.5
	IVL2.5 [11]	18.2	48.7	26.5	32.3	36.1	30.1	34.2
	QVL2.5 [3]	16.5	50.5	24.8	34.1	29.1	30.5	32.0
	GFT* [20]	36.5	47.2	41.1	50.1	43.2	33.6	44.5
Exo	VL3 [62]	20.1	44.2	27.6	34.7	34.8	31.2	29.1
	IVL2.5 [11]	18.7	48.8	27.0	37.5	33.3	29.8	32.5
	QVL2.5 [3]	21.1	49.6	29.6	32.4	29.4	31.4	32.6
	GFT* [20]	35.1	51.1	41.6	48.5	41.0	34.3	46.6

Table 4: Zero-shot evaluation for MD on the first 15 tasks. **Top:** Egocentric Perspective. **Bottom:** Exocentric Perspective. A similar trend is seen across the two views.

the correct action or a mistake. For the test, users designed specific prompts that instructed the VLM to look for specific factors (e.g. *did the user use the appropriate tool for the task?*), and also to check if a known mistake is occurring (e.g. *did the user leave the trolley in the corridor?*). The cluttered background and industry-specific nature of the data make it difficult for the VLMs to accurately reason about the action.

For finetuning results, we extract the feature embeddings for the action steps from the VLMs, which are then forwarded to either an MLP classifier or a Transformer (Tr) based classifier. Steps with *mistakes* are assigned a higher weight (5). For early mistake detection (EM), only the initial 50% of frames are available to the model during validation/testing. Table 4 compares the zero-shot evaluation results on the first 15 MD tasks for the egocentric and exocentric perspectives.

We evaluated mistake detection on a subset of 10 tasks to assess the value of joint ego- and exocentric views, reporting F1 scores in Table 5. Our study included zero-shot (ZS) evaluation and a finetuning approach where we extracted step-wise embeddings from a VLM, concatenated the corresponding view embeddings, and trained a transformer-based model (+Tr) with k-fold cross-validation. The joint Ego + Exo view slightly outperforms single-view inputs across all models. A qualitative review of the zero-shot results indicates the exo view is most beneficial

Model	Ego	Exo	Ego + Exo
GFT (ZS) [20]	0.43	0.39	0.44
VL3 + Tr [62]	0.33	0.32	0.37
IVL2.5 + Tr [11]	0.29	0.30	0.33

Table 5: Mistake detection F1 scores for Ego-only, Exo-only and joint-views. We compare zero-shot (ZS) and finetuned (+Tr denotes a transformer block) models. The joint view consistently yields the best performance.

when the ego view is occluded or the mistake is out of focus, though in some cases it offered no improvement or led to incorrect predictions.

Ablation Study on Modality Contribution. We conducted an experiment on the mistake detection portion of the dataset (10 tasks across all scenarios), reporting average F1 scores in Table 6. For zero-shot evaluations, eye gaze was overlaid on the RGB stream, and the model was prompted to focus on the indicated region. Our qualitative review reveals that a modality’s impact is context-dependent. For instance, removing the audio modality in noisy environments can slightly improve performance. Conversely, in collaborative tasks, audio provides a crucial signal for understanding context. Similarly, prompting the model to use eye gaze is beneficial when a user performs an action incorrectly within their field of view. However, when the mistake occurs outside the gaze region (e.g., forgetting to pack an object in a container), adding the gaze modality can slightly worsen performance.

Model	RGB only	RGB + Audio	RGB + Gaze	RGB + Audio + Gaze
GFT (ZS) [20]	0.38	0.41	0.39	0.42
VL3 [62]	0.27	0.26	0.28	0.30
IVL2.5 [11]	0.30	0.28	0.29	0.29

Table 6: F1 scores for *Mistake Detection* (averaged across 10 tasks), ablating audio and gaze modalities. Our investigation into the results for each *task* shows that the utility of each modality is highly context-dependent.

4.2 Reasoning-Based Video Question Answering

It is essential for egocentric assistants and agents to be able to understand the scene, the task, and reason about the implications of various actions. Current state-of-the-art (SOTA) multimodal models have demonstrated remarkable ability w.r.t. processing text and image data [7, 32, 42, 57]. There has also been an increased focus on reasoning and planning [41, 54, 61]. Recent Video Language Models (VLMs) have been shown to understand allocentric videos, summarise and answer questions [29]. These capabilities also generalise to egocentric and industrial scenarios. We curated a set of 3105 question and answer pairs, based on the *singular actions*, *fine-grained annotations* (2020) and the other categories (1085).

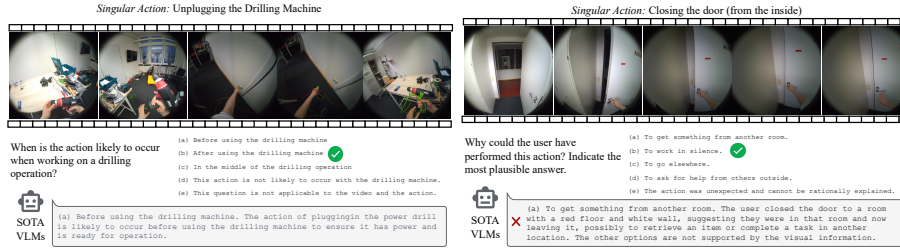


Fig. 6: Example questions from the reasoning-based QA benchmark from the dataset, followed by an answer produced by a SOTA model [62]. **Left:** The Answer was wrong, potentially because the model misunderstood the temporal action (attaching vs. detaching). **Right:** The answer is wrong because of incorrect situated reasoning (closing the door from the inside is different from walking out and closing the door).

The questions require visual perception and reasoning about the user’s action and the objects in the field of view. The questions are designed so that an average technician (or a layperson with access to the internet) can answer them. Figure 6 shows an example of two questions based on the *singular actions*. The questions can be broadly split into four groups: those focusing on the temporal aspect of the action (Tm - 14%), those requiring situated reasoning (Si - 28%), those requiring visual recognition of the objects in the FOV (Re - 32%), and those requiring analogical or abductive reasoning (A - 26%).

Table 7 gives the result (% accuracy) for SOTA VLMs on the *singular actions* portion of the benchmark. As seen in Figure 6, questions that can be readily answered by humans, could be challenging for VLMs. For comparison, the questions were also answered by human evaluators (4).

Model	Acc Tm	Acc ^{Si}	Acc ^{Re}	Acc ^A	Acc
VL3 [62]	52.2	60.3	59.4	57.5	58.2
IVL2.5 [11]	51.7	61.1	58.2	56.0	57.6
QVL2.5 [3]	53.2	60.8	59.3	56.5	58.1
GFT* [20]	55.4	62.1	67.2	68.3	64.1
ML2* [39] + Label	92.3	51.4	42.8	78.3	61.4
Human	92.6	89.6	90.4	88.6	90.0

Table 7: Zero-shot evaluation of the reasoning based QA benchmark on singular actions portion of the dataset with some SOTA VLMs. Mistral-Large2 model was given the action label (e.g. closing the door from the inside) and asked to select a correct answer. Human evaluators had access to the industrial tool database from the dataset. *via API, Experimental, model architecture unspecified.

4.3 Task Understanding in a Collaborative Setting

It is important for egocentric assistants to understand the actions of the wearer, as well as others, esp. when working together on a common task. We provide

extracted action pairs from the dataset for procedural and non-procedural tasks. The goal is to differentiate and predict the actions of the user and the coworker for the given segment, and understand their relative role (e.g. collaborator, teacher/student). A zero-shot evaluation yields an accuracy of 35.2% (GFT), with the VLM struggling to differentiate between the egocentric viewer’s actions and the actions of the coworker. Employing a similar setup as MD (VLM embeddings forwarded to a classifier). We train the model to predict what the wearer would do based on the actions of the coworker, and report a baseline accuracy of 42.1% (VL3 + Tr). Both evaluations were performed on approximately 50% of the collaborative egocentric recordings from each scenario. Ablation studies show, that removing the audio modality negatively impacts the zero-shot results.

4.4 Additional Experiments

Summarisation, Long Video Understanding and Reasoning-QA. The goal is to produce a coherent summary of the input video (text and image), with minimal loss in context and key events. As ground truth, we sample the egocentric recordings based on the keysteps and fine-grained annotations. The dataset contains up to 68 min long videos. The longer videos tend to belong to the *unguided* sequences (e.g. the user is given a repair task without any instructions). These scenarios reflect the real-world cases where an egocentric agent could be beneficial. The reasoning-based QA task for the longer tasks (1085 questions) is designed to assess this aspect. Audio-only (transcript from user narration) yields a baseline accuracy of 67.3%. Additionally, we provide summaries and context on the tasks performed by the participants, along with additional comments (e.g. unplanned mistakes, incorrect practices). For collaborative tasks, we provide summaries from both points of view and an overall summary.

Additional Modalities. Certain tasks and actions are inherently multimodal (e.g. attaching a battery to the drill produces a *click* sound). Additionally, other modalities (user trajectory, gaze) and the exocentric perspective offer a rich array of research possibilities, including cross-view and cross-modal alignment.

5 Conclusion

We introduced *IndEgo*, a unique multimodal dataset for egocentric and exocentric vision in industrial tasks, including assembly/disassembly, logistics organisation, inspection, repair, and woodworking. IndEgo consists of 3,460 egocentric recordings (approximately 197 hours) and 1,092 exocentric recordings (approximately 97 hours), along with eye gaze, narration, audio, motion, and other sensor data. We provide detailed annotations and challenging benchmarks for procedural and non-procedural task understanding, Mistake Detection, and reasoning-based Question Answering. Baseline results on Mistake Detection, Question Answering and collaborative task understanding show that the dataset presents a challenge for the SOTA multimodal models, highlighting the need for further exploration. IndEgo consists of diverse cognitively and physically intensive industrial tasks,

filling a relevant gap in egocentric vision research. We believe it will be a valuable resource for advancing AI-assisted industrial applications and enhancing productivity, safety, and efficiency in real-world operations.

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